# MICROBIOLOGY

CHAPTER 6

Viruses and Other Acellular Infectious Agents

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**Reference:**

**Slides, Dr.Sereen records & the book**

# Chapter 6: Viruses and Other Acellular Infectious Agents

We're going to learn about: (–‿‿–)♡ (´｡• ᵕ •｡`) ♡

1. Viruses: Definition and Structure
2. Viral life cycles
3. Types of viral infections
4. Cultivation & Enumeration of viruses
5. Viroids , Satellites & Proins

Acellular Agents:
• Viruses →protein and nucleic acid
• Viroids →only RNA
• Satellites →only nucleic acids
• Prions →proteins only

(كل هدول لا يُعتبروا كائنات حية لعدم تكونهم من وحدة الحياة الاساسية : الخلية)

## ❁ Viruses: Definition and Structure

- Viruses: Acellular infectious agents, simple and contain either DNA or RNA (NOT both)

الخلية بكون عندها دائما DNA وRNA مع بعض (يعني بتضل تصنع RNA من الDNA) وبتقدر تتكاثر وتتضاعف لحالها
..الvirus بالمقابل ما عندو هاي الصفات وبحتاج خلية عشان يتضاعف فيها ويستحوذ على آلياتها ومنحكيلو intracellular parasite لانو بدخل الخلية وبتضاعف هناك

- Roles of viruses:(أو ليه من المهم ندرس الفايروسات)
  - Major cause of disease
    →also importance as a new source of therapy
    →new viruses are emerging
  - Important members of aquatic world
    → move organic matter from particulate to dissolved
  - Important in evolution
    →transfer genes between bacteria, others
  - Important model systems in molecular biology
- General Properties of Viruses
  - **Virion** : a complete virus particle
    →consists of ≥1 molecule of DNA/RNA in protein coat
    →may have additional layers
    →cannot reproduce independent of living cells nor carry out cell division
    BUT can exist extracellularly

الvirion هو جزيء الفايروس الكامل المكوّن من على الاقل جزيء DNA اوRNA واحد مع غطاء بروتين من برا وممكن يحتوي على طبقات تانية وبسبب تركيبه البسيط ما بقدر يتضاعف لحالو..لازم يكون في خلية مضيفة لتساعده بس ممكن يعيش ويتواجد لحالو برا الخلايا حية حسب الenvironmental conditions

- Virions can infect all cell types
  - Bacterial virions are called Bacteriophages (or phages for short)
  - They can infect Archaea
  - Most are Eukaryotic viruses:
    →plants, animals, protists, and fungi

الفايروسات بتقدر تسبب العدوى لأي كائن حي..سواء كان بكتيريا او حيوان او فطريات او نباتات ..كلها قابلة للعدوى ولكن الفايروس الي مثلا بعدي البكتيريا ما بقدر يعدي اي شيء تاني لهيك منحكيلها انها highly specific..حتى الي بتعدي الرئة ما بتقدر تعدي الكبد وهاد كله بسبب وجود مستقبلات(receptors) محددة ومخصوصة على سطح الخلية الي بعديها لازم تكون موجودة عشان يسبب العدوى

  - The classification of viruses is based on:
    Genome structure, Life cycle, Morphology, Genetic relatedness

- Structure:
  - Virion size range is ~10–400 nm in diameter and most viruses must be viewed with an electron microscope
  - All virions contain a **nucleocapsid** which is composed of nucleic acid (DNA or RNA) and a protein coat (capsid) – some viruses consist only of a nucleocapsid, others have additional components

يعني يا بكون الفايروس بس عبارة عن nucleocapsid( nucleo → DNA or RNA || capsid→ protein coat) او ممكن يكون بملك كمان تركيبات ومكونات غيرهم

  - **Envelope**: a lipid layer external to the nucleocapsid

الenvelope ما الو اي علاقة بحماية المادة الوراثية او حماية الفايروس هو بس زي الكيس الي بِلم كل شي مع بعضو See figure 6.2

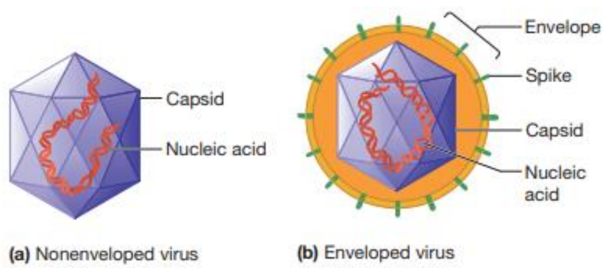



**Figure 6.2 Generalized Structure of Virions.** (a) The simplest virion is that of a nonenveloped virus (nucleocapsid), consisting of a capsid assembled around its nucleic acid. (b) Virions of enveloped viruses are composed of a nucleocapsid surrounded by a membrane called an envelope. The envelope usually has viral proteins called spikes inserted into it.

→Virion structure is defined by: Capsid Symmetry and Presence or Absence of an Envelope

A. Capsids:
  • Large macromolecular structures which serve as protein coat of virus
  • Function: Protect viral genetic material and aids in its transfer between host cells
  • Made of protein subunits called **Protomers** :they are the subunits of capsids and they need a host cell to be made but they can assemble by themselves (self-assembly)

- Three types of capsid symmetry:

a. Helical:
   - Shaped like hollow(مفرّغة) tubes with protein walls
   - Size of capsid is a function of nucleic acid
   - Examples:

     -Tobacco Mosaic Virus (TMV)→ Nonenveloped

     -Influenza Virus→ Enveloped

     →consist of 7 to 8 different RNA molecules, each is coated by capsid proteins to create 7 to 8 flexible nucleocapsids

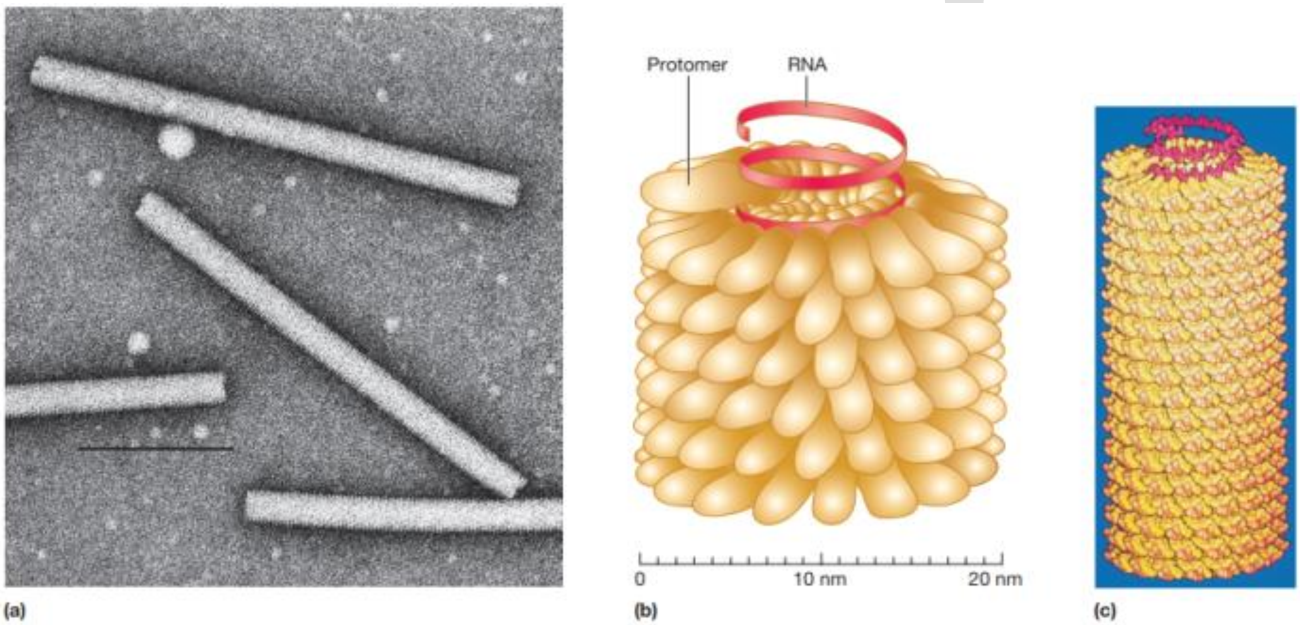
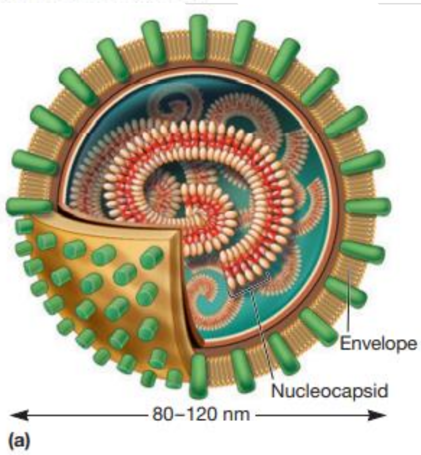



**Figure 6.3 Tobacco Mosaic Virus (TMV) Virions.** (a) An electron micrograph of negatively stained helical capsids (×400,000). The virions are usually 15 to 18 nm in diameter and 300 nm long. (b) Illustration of a TMV nucleocapsid. Note that it is composed of a helical array of protomers with the RNA spiraling on the inside. (c) A model of a TMV virion.



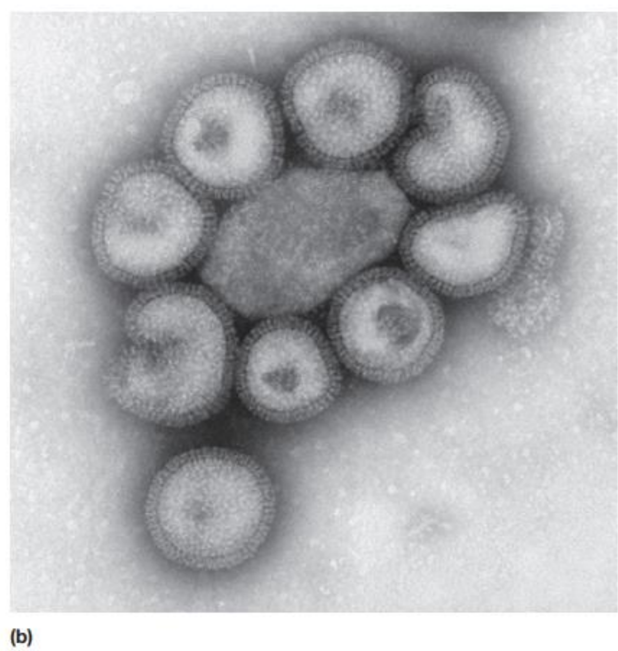




**Figure 6.4 Influenza Virus Virions: An Enveloped Virus with Helical Nucleocapsids.** (a) Schematic view. Influenza viruses have segmented genomes consisting of seven to eight different RNA molecules. Each is coated by capsid proteins to create seven to eight flexible nucleocapsids. The envelope has spikes that project 10 nm from the surface at 7 to 8 nm intervals. (b) Because the nucleocapsids are flexible, the virions are pleomorphic. Electron micrograph (×350,000).

Polyhedron :a solid figure with many plane faces, typically more than six.

Icosahedral Capsids

- An icosahedron is a regular polyhedron* with 20 equilateral faces and 12 vertices (peaks)
- Capsomers – ring or knob-shaped units made of 5 or 6 <u>protomers</u>

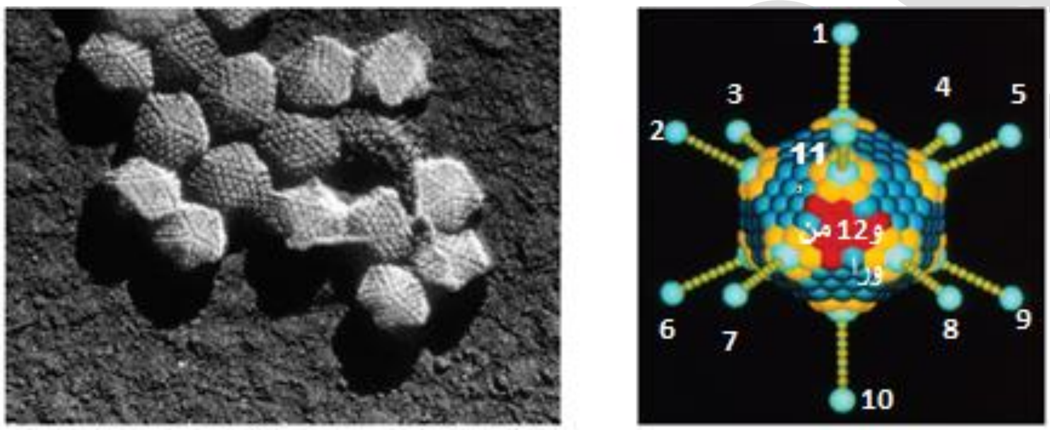
  – pentamers (pentons) → 5 subunit capsomers

  – hexamers (hexons) → 6 subunit capsomers

الcapsomer هو مجموعة من5-6 protomers متجمعين مع بعض على شكل حلقة.. وردة او المقبض

لما يتجمعوا 5 capsomers مع بعض بسمي التركيب Pentamer او Pentons

ولما يتجمعوا 6 capsomers مع بعض بسمي التركيب Hexamer او Hexons




(a) (b)

**Figure 6.5** **The Icosahedral Capsid of an Adenovirus.** (a) Electron micrograph of adenovirus virions, 252 capsomers (×300,000). (b) Computer-simulated model of an adenovirus virion.

c. Complex

- Some viruses do not fit into the category of having helical or icosahedral capsids

  ( لا همة helical ولاicosahedral وما الهم شكل محدد)

- Examples:

  - Poxviruses → largest <u>animal</u> virus

    →double stranded DNA genome is associated with proteins and contained in the core

    →it has a central structure shaped like a biconcave disk and surrounded by a membrane.

    →Two lateral bodies lie between the core and the virion's outer envelope.

  -Large bacteriophages →binal symmetry

    - head → icosahedral (have the nucleic acid)

    -tail → helical

    -Has a Sheath that connects the head with tail fibers

    ⇨Looks like a Space ship

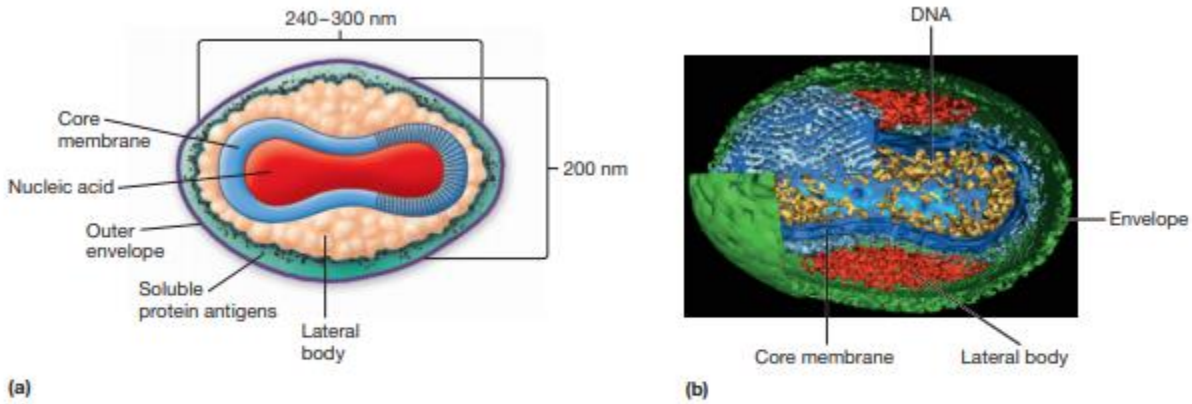



**Figure 6.6 Morphology of Vaccinia Virus Virions.** (a) Diagram of virion structure. (b) Electron cryotomogram of the virion.

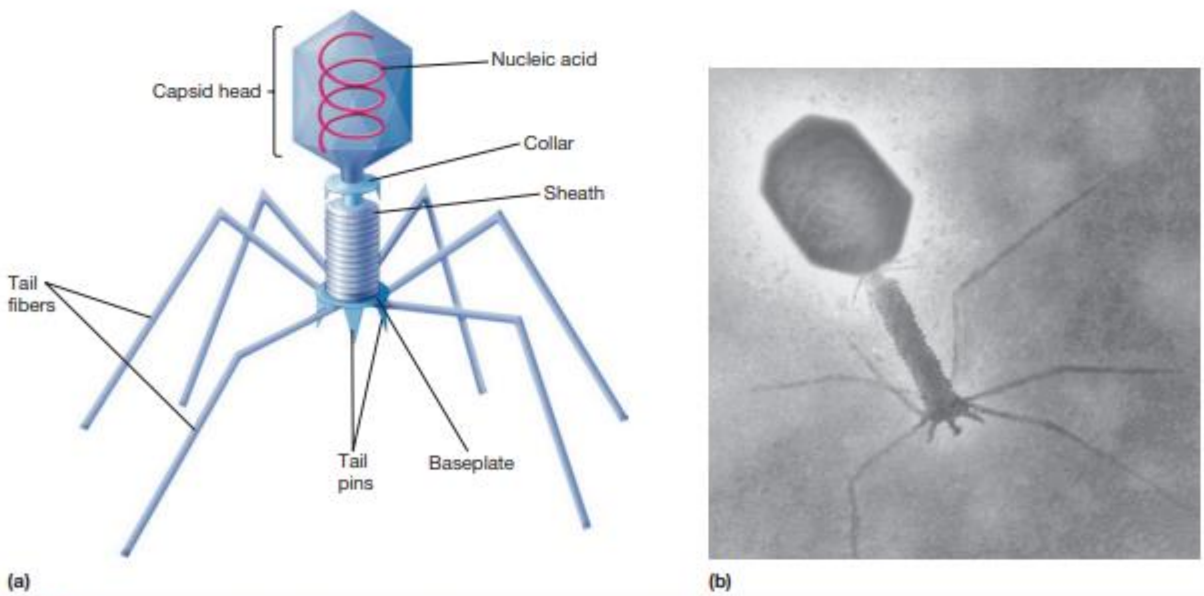



**Figure 6.7 T4 Phage of *E. coli*.** (a) The structure of T4 bacteriophage virion. (b) Electron micrograph of the virion.

B. Envelopes:

- an outer, flexible, membranous lipid layer

Two types of viruses according to the existence of envelopes:

1- Non enveloped (Naked virus): Only a nucleic acid surrounded by a capsid

2- Enveloped :has an envelope around the nucleocapsid

- Envelope proteins, which are viral encoded (يعني المادة الوراثية تبعت الفايروس نفسه هي الي بتحدد تكوينهم), may project from the envelope surface as **Spikes** or peplomers

  They have 3 main important functions:

  1- involved in <u>Viral Attachment</u> to host cell
     →• e.g., **hemagglutinin** of influenza virus
  2- may have enzymatic or other activity
     →e.g., **neuraminidase** of influenza virus
  3- used for identification of virus
  4- may play a role in nucleic acid replication

فايروس الانفلونزا عندو نوعين من البروتينات : **Hemagglutinin & Neuraminidase** على الenvelope تبعو ..بفرّقوا بين انواع الفايروسات الي بتسبب الانفلونزا عن طريق اعطاء هاي البروتينات ارقام ..مثلا،الفايروس الي بسبب انفلونزا الطيور اسمو H5N1 بينما انفلونزا الخنازير اسمو H1N1

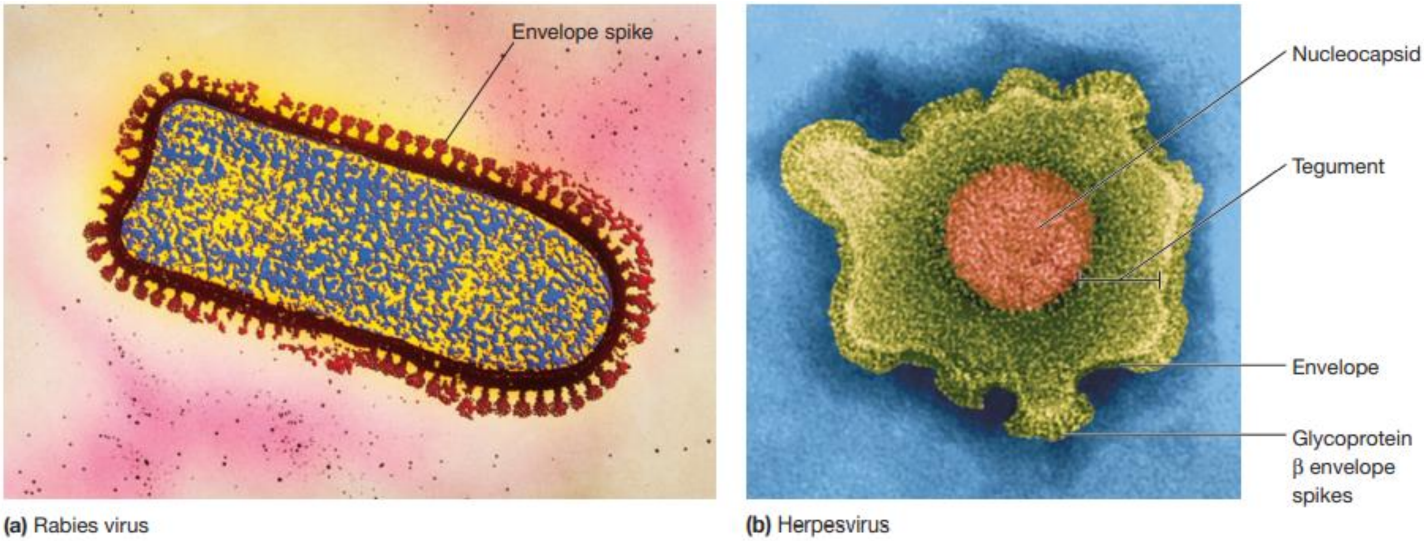



**Figure 6.8 Examples of Enveloped Viruses.** (a) Negatively stained virion of a rabies virus. (b) Herpesvirus virion. Images are artificially colorized.

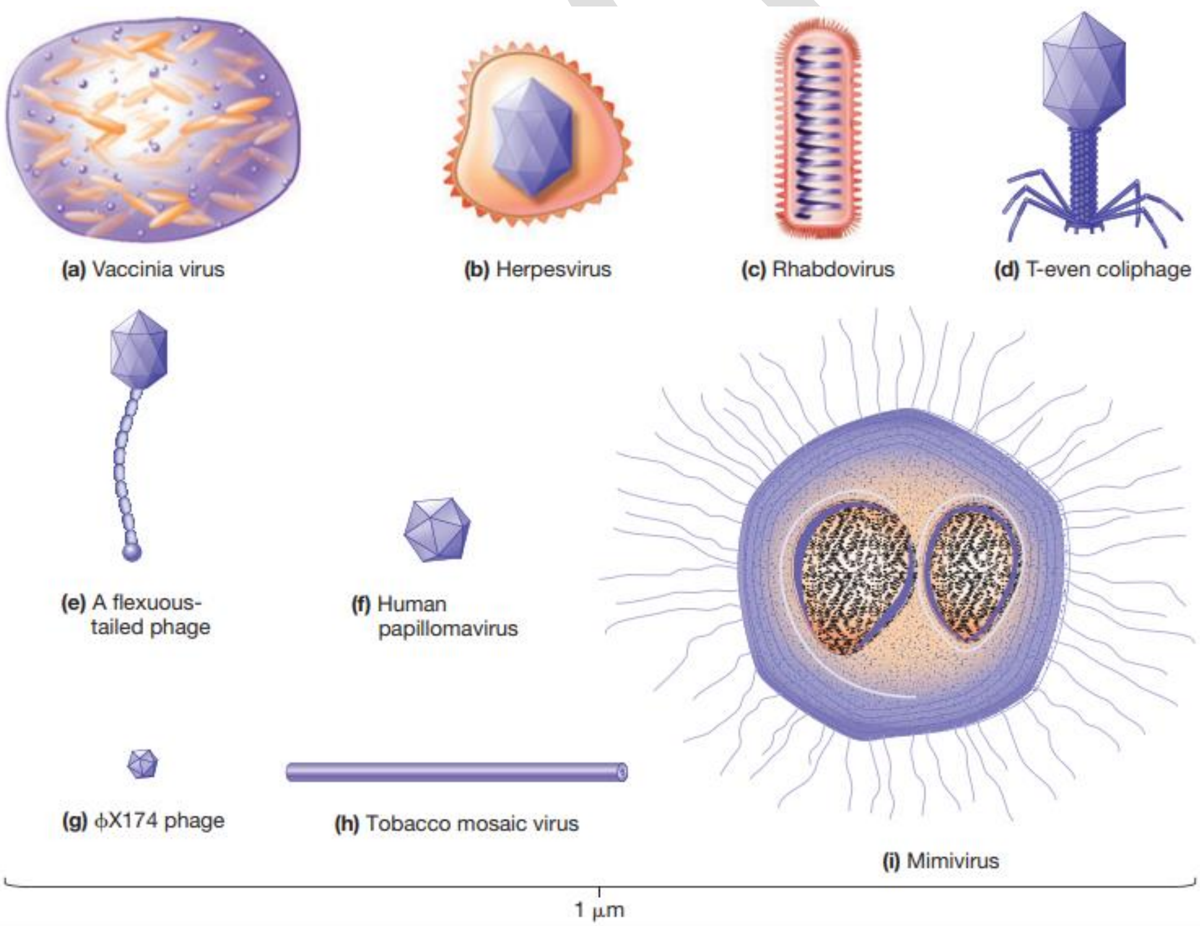



**Figure 6.1 The Size and Virion Morphology of Selected Viruses.** The virions are drawn to scale.

- Virion Enzymes
  - First erroneously thought that all virions lacked enzymes (ↄ´ω`)/(╥ω╥)
    Now accepted that a variety of virions have enzymes – some are associated with the envelope or capsid but most are within the capsid
- Viral Genome
  - Diverse nature of genomes→a virus may have single or double stranded DNA <u>or</u> RNA
  - The length of the nucleic acid also varies from virus to virus
  - Genomes can be segmented or circular

المادة الوراثية للفايروسات جدا متنوعة ..ممكن تكون double او RNA single او double او DNA single وممكن تكون حلقية (دائرية) او على شكل قطع بالإضافة الى انو طول المادة الوراثية مو ثابت وبفرق من فايروس لفايروس

## Viral life cycles

- One-Step Growth Curves

Delbruck and Ellis did experiments with bacteriophage T4 and *E. coli* host cells in 1939 where they knew that T4 can kill *E.coli* since *E.coli* is its host and after infection, progeny phages were released because they lysed the cell they infected
→this experiment was the first step and evidence to understanding viral life cycle steps
Steps of the experiment:

1. They mixed T4 and *E.coli*
2. After a short interval, the mixture was greatly diluted so that any virions released upon cell lysis would not be able to infect other cells
3. The diluted culture was then incubated and over time samples were taken to determine the number of infectious phage particles in the culture
4. A plot of phage particles versus time shows several distinct periods in the resulting curve →figure 6.9

التجربة باختصار:

Delbruck وEllis خلطوا فايروس الT4 والE.coli مع بعض وبعد فترة من الزمن خففوا المحلول كتير عشان اي فايروس جديد بنتج ما يقدر يعدي اي خلية تانية(يعني عدد الخلايا قلت عشان الفايروسات الجديدة ما تعديها ويرجع يتضاعف العدد) بعدين عملوا incubation للمحلول وضلوا ياخدوا منو عينات عشان يعدوا كم فايروس في بالمحلول كل ما يمر الوقت وحطوا نتائجهم برسم بياني زي ما هو موضح بالشكل والهدف منها كلها اني اعرف شو المراحل الي بمر فيها الفايروس من لما يسبب infection لحتى يطلعوا الفيروسات السنافير الجداد

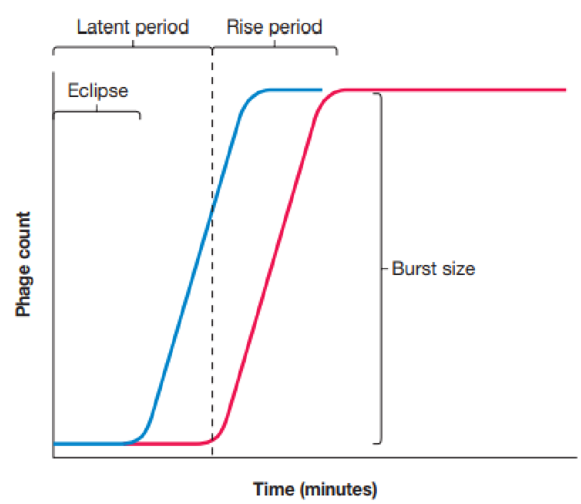



**Figure 6.9** **The One-Step Growth Curve.** The eclipse period is the initial portion of the latent period in which host cells do not contain any complete, infective virions. During the remainder of the latent period, an increasing number of infective virions are present, but none are released. The latent period ends with host cell lysis and rapid release of virions (rise period). In this figure the blue line represents the total number of complete virions , both intracellular and free in the medium. The red line represents the number of free viruses (virions that were free in the culture and had not infected a host cell plus those virions that were released from infected cells). The burst size is the number of progeny viruses produced per infected cell.

Results:

- **Eclipse**: when phages enter and disappear inside cells causing infection. It is called "eclipse" because the infectious virions are concealed or eclipsed within host cells
- **Latent period:** occurs immediately following the addition of phages. During this period, NO virions are released
- **Rise period**: cells are lysed and virions are released. It is characterized by the Rapid release of infectious phages

  →Finally, a plateau is reached where no viruses are released

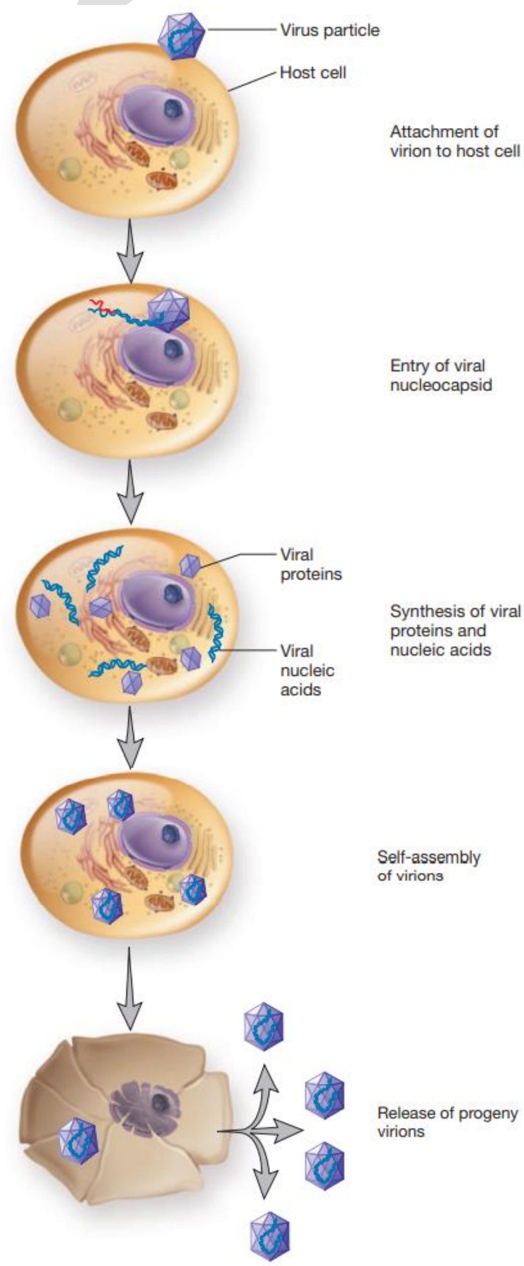



**Figure 6.10 Generalized Illustration of Virus Multiplication.** The life cycle of a virus that infects a eukaryotic cell is shown. Figure 6.16 illustrates the generalized life cycle of many bacterial and archaeal viruses. Virions are not drawn to scale.

- Viral Multiplication
  - The mechanism used depends on viral structure and genome

    But Steps are similar:

    1) Attachment to host cell
    2) Entry and uncoating of genome
    3) Synthesis
    4) Assembly
    5) Release

1) Attachment (Adsorption)
   - The most important step for a virus to cause infection and multiply

     →if we manage to prevent the attachment of the virus, we will never get infected
   - Specific receptor attachment:

كل الفايروسات (ما عدا النباتية) لازم ترتبط مع الخلية المضيفة لوقت كافي عشان تدخل الخلية. الترابط هاد بتحقق بسبب وجود تفاعلات ما بين جزيئات على سطح الvirion منسميها ligands مع جزيئات على سطح الخلية منسميها receptors .

الreceptors والligands متل المفتاح وقفله..كتير specific ومخصصة ومو أي ligand برتبط مع اي receptor

   - Receptor determines host preference
     - may be specific tissue (tropism)
     - may be more than one host
     - may be more than one receptor
     - may be in lipid rafts providing entry of virus

الreceptors الي على سطح الخلية المضيفة ممكن يكونوا:

-cell wall lipopolysaccharides and proteins

-teichoic acids, flagella, or pili

لهيك الفايروس بكون بعدي نوع نسيج معين او ممكن يكون الو اكتر من host وممكن يكون الligand تبعو بركب على اكتر من receptor ف في احتمالات كتيرة للموضوع ..بلس انو عندي مناطق بالmembrane تبع الhost cell منسميها lipid rafts ممكن يساهموا في عملية الارتباط (attachment) والدخول(entry) تبع الفايروس

2) Viral Entry and Uncoating

- May be achieved by the injection of the viral genetic material while the capsid remains outside or the entry of the entire nucleocapsid
- <u>Varies between naked or enveloped virus</u>
- Three methods used –
  - Fusion of the viral envelope with host membrane→nucleocapsid enters

الenvelope تبع الفايروس بندمج مع غشاء الخلية المضيفة وبدخل الnucleocapsid لحالو figure 6.11 a

  - Endocytosis in vesicle; endosome aids in viral uncoating

الفايروس كامل بدخل الخلية عن طريق الendocytosis او البلعمة. الendocytic vesicle (الي هي عبارة عن الفايروس والغشاء الي فات معه من غشاء الخلية نفسو) بتدخل جوا ال endosome ..بناءا على نوع الفايروس، يا اما الnucleocapsid بطلع لحالو برا الendosome أو المادة الوراثية تبعت الفايروس بتطلع من ال endocytic vesicle لحالها..يا اما قبل ما تدخل جوا الendosome او بعد. في انزيمات جوا الendosome ممكن انها تساعد بعملية الuncoating للفايروس(الي هي نفسها خروج المادة الوراثية من الcapsid).

الفرق بين الenveloped والnonenveloped بكيفية الخروج من الendosome انو الenvelope تبع الenveloped viruses بندمج مع الغشاء تبع الendosome والnucleocapsid بطلع. اما بحالة الnonenveloped، الPH القليلة تبعت الendosome بتسبب conformational change بالcapsid ويا إمّا بتطلع المادة الوراثية لحالها او كل الnucleocapsid

  - Injection of nucleic acid

بقدر الفايروس كمان انو يحقن المادة الوراثية لحالها جوا الخلية ويضل الcapsid برا

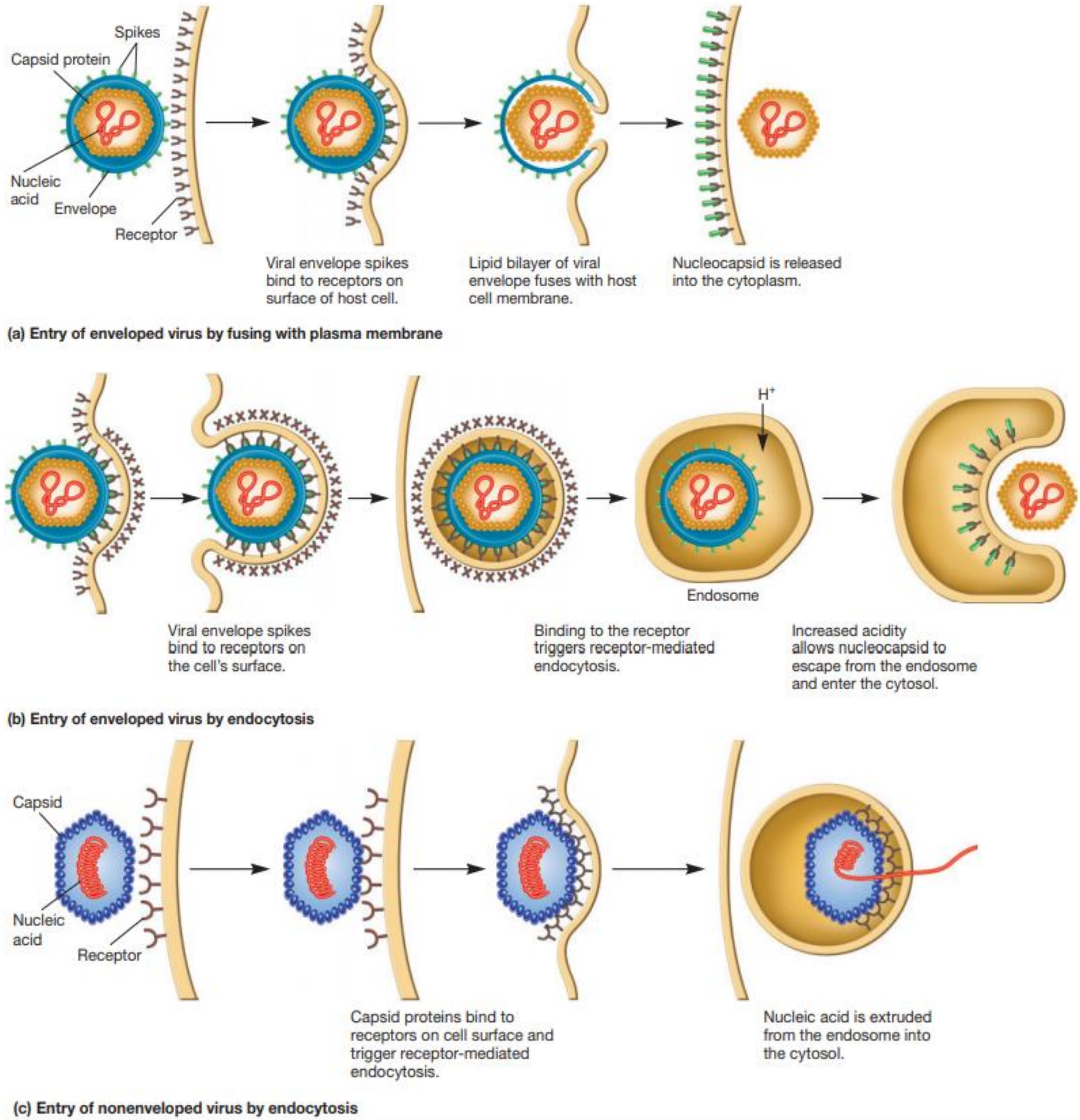



**Figure 6.11 Animal Virus Entry.** Examples of animal virus attachment and entry into host cells. Enveloped viruses can (a) enter after fusion of the envelope with the plasma membrane or (b) escape from an endosome after endocytosis. (c) Nonenveloped viruses may be taken up by endocytosis and then insert their nucleic acid into the cytosol through the vesicle membrane. It also is possible that they insert the nucleic acid directly through the plasma membrane.

3) Synthesis Stage :

- In this stage, the virus shuts down the metabolism of the host cell and takes over the host cell DNA to make its own parts (proteins and nucleic acid) (the cell becomes a virus factory)
- This stage of the viral life cycle differs dramatically among viruses because <u>the genome of a virus</u> dictates the events that occur

  →For ds DNA viruses, the synthesis stage can be very similar to the typical flow of information in cells

الtypical flow الي هو انو المعلومات المحفوظة بالDNA بصير نقلها للmRNA ومن الmRNA للرايبوسومات عشان يتم صنع البروتينات← عملية الtranslation الي كلنا منعرفها

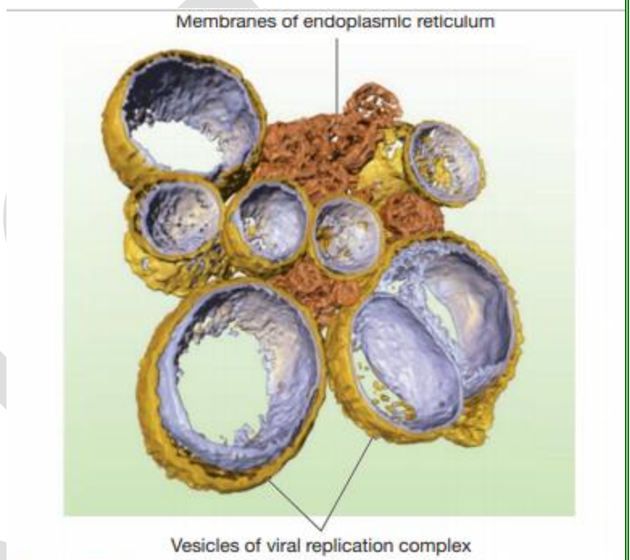



**Figure 6.12 A Viral Replication Complex.** An electron tomogram of cells infected with severe acute respiratory syndrome coronavirus (SARS-CoV). The vesicles of the viral replication complex have a double membrane derived from endoplasmic reticulum (ER) membranes. The membranes of vesicles are connected to the ER.

  →RNA viruses: virus must carry in or synthesize the proteins necessary to complete synthesis

الRNA viruses ما فيهم الانزيمات اللازمة لنسخ المعلومات عن الRNA او لترجمتها لmRNA لهيك لازم يكون عندهم كل الانزيمات اللازمة لإنهاء الsynthesis stage بالnucleocapsid تبعهم او انو يتم تصنيعها خلال الinfection process

- Stages may occur, e.g., early and late

من الصفات المهمة للsynthesis stage انها منظمة بشكل كبير . بحيث يتم تقسيم الجينات لearly وmiddle وlate حسب متى بصيرلها expression والبروتينات الي بعطوها برضو بكونوا early او middle او late

- May induce formation of membrane-protected replication complexes

ممكن الفايروسات تستولي على الmembrane system (Golgi, ER,..) تبع الخلية المضيفة وتعمل تركيبات مكونة من اغشية عشان تحمي نفسها والعمليات الي بتعملها من اليات الدفاع تبعون الخلية المضيفة ،هاد التركيب منسميه **viral replication complex** ←figure6.12

4) Assembly

- Viruses self-assemble meaning that the cell has nothing to do with it→capsid protomers themselves have the information to organize and assemble the parts of the virus
- Late proteins are important in assembly

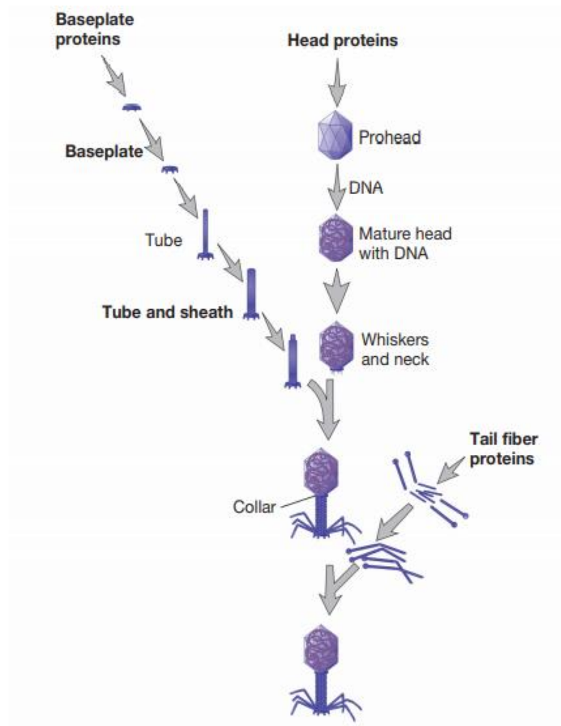



**Figure 6.13 The Assembly of T4 Virions.** Note the subassembly lines for the baseplate, tail tube and sheath, tail fibers, and head.

- Assembly is complicated but varies
  → bacteriophages – stages
  →some are assembled in nucleus
  →some are assembled in cytoplasm
  →may be seen as paracrystalline structures in cell

5) Virion Release

- Nonenveloped viruses lyse the host cell and leave
  - viral proteins may attack peptidoglycan or membrane
- Enveloped viruses use <u>budding</u>
  1. First, virus-encoded proteins are made by the host cell and incorporated into the membrane
  2. nucleocapsid may bind to viral proteins
     - envelope derived from host cell membrane, but may be Golgi, ER, or other
     - virus may use host actin tails to propel through host membrane in order to not be recognized by the immune system

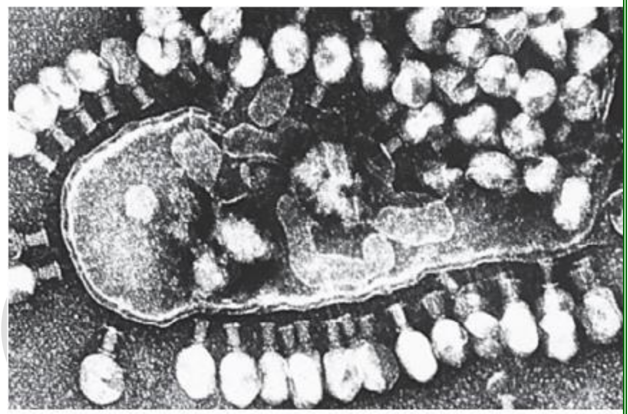


**Figure 6.14** **Release of T4 Virions by Lysis of the Host Cell.** The host cell has been lysed (upper right portion of the cell) and virions have been released into the surroundings. Progeny virions also can be seen in the cytoplasm. In addition, empty capsids of the infecting virus particles coat the outside of the cell (×36,500).

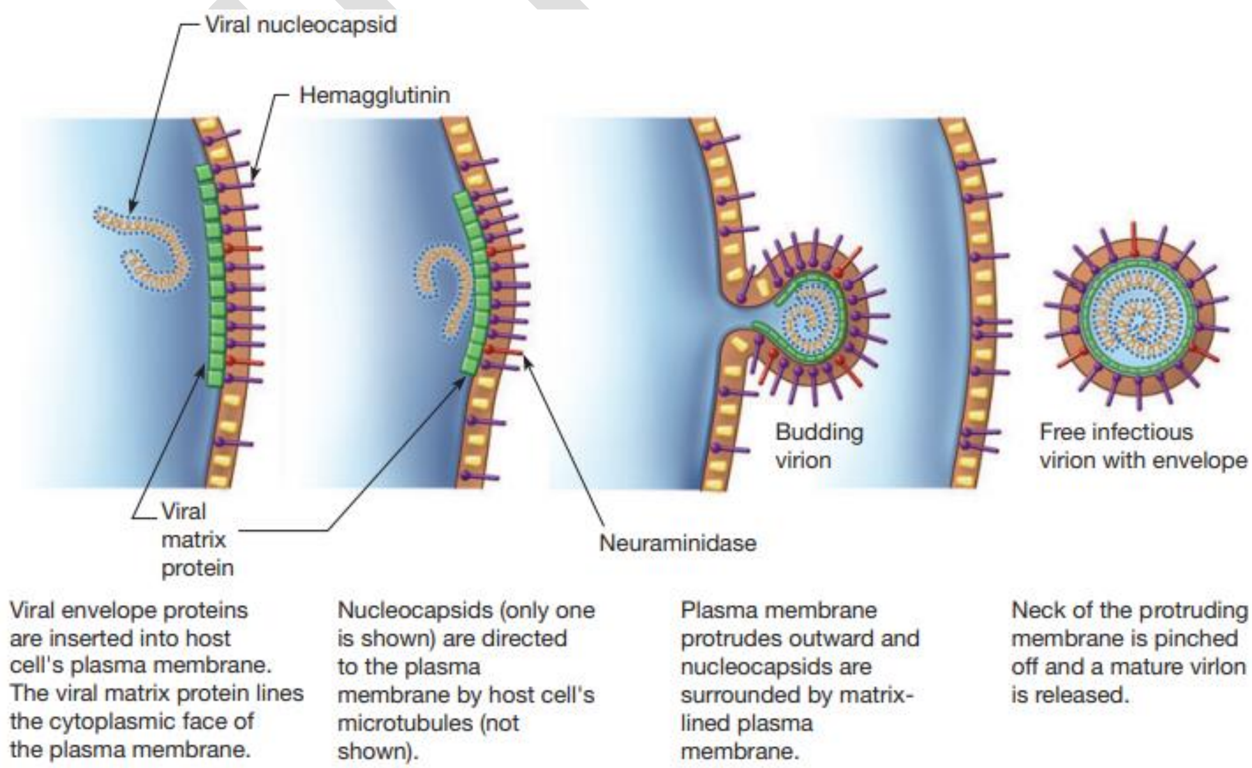



**Figure 6.15** **Release of Influenza Virus Virions by Budding.** For simplicity, only one of the seven to eight possible nucleocapsids are shown.

## ❀ Types of viral infections

- There Are Several Types of Viral Infections:

A) Bacterial and Archaeal Viral Infections

- Different types of phages:

a. Virulent phages→ they have one reproductive choice
They multiply immediately upon entry and then lyse bacterial host cell and leave

b. Temperate phages→have two reproductive options
⇨Either they reproduce **lytically** as virulent phages do→**Lytic cycle**
⇨OR the virus enters its genetic material to the host cell and then they integrate it with the host cell genetic material; they remain there latent and does not reproduce or lyse the bacteria →**Lysogenic cycle**
but sometimes and for certain reasons, they convert the lysogenic cycle to lytic cycle

- ❖ When the virus integrates its genetic material with the genetic material of the host cell and the virus remains latent, the virus is called **Prophage** and the bacterial cell is called **lysogenic bacterium**
**Figure 6.16**

➔ Lysogenic Conversion : occurs when a temperate phage changes the phenotype of its host – bacteria become immune to superinfection – phage may express pathogenic toxin or enzyme

مثال على هاي العملية الي بصير بال*Salmonella* لما تنعدي من ال**epsilon phage**. الفايروس هاد بغير بوظائف العديد من الانزيمات الي الها علاقة ببناء المكون المصنوع من الكاربوهيدرات بال **lipopolysaccharides** تبعت البكتيريا ..هاد بزيل ال**receptors** تبعت هاد الفايروس نفسه من على البكتيريا عشان ما تنعدي من اي واحد تاني غيره (يعني زي كأنها بصير عندها مناعة ضد هاد الفايروس)..الي صار مع ال**salmonella** يُعتبر **phenotypic change**

• Two advantages to lysogeny for virus :
– phage remains viable but may not replicate
– multiplicity of infection ensures survival of host cell
• Under appropriate conditions infected bacteria will lyse and release phage particles and this occurs when conditions in the cell cause the prophage to initiate synthesis of new phage particles→a process called **induction**

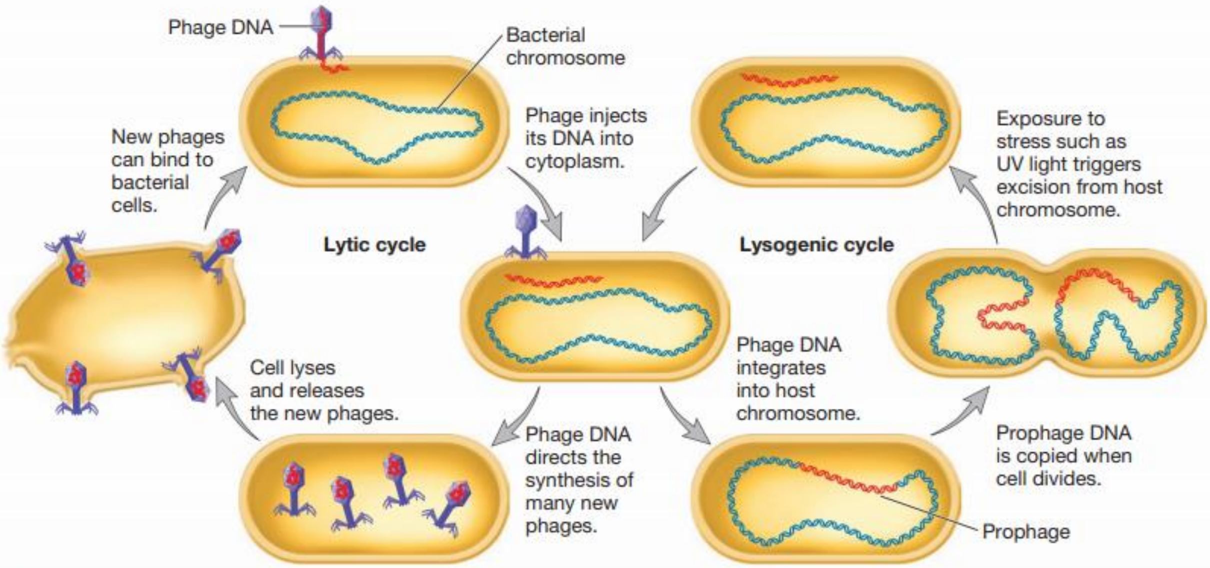



**Figure 6.16** **Lytic and Lysogenic Cycles of Temperate Phages.** Temperate phages have two phases to their life cycles. The lysogenic cycle allows the genome of the virus to be replicated passively as the host cell's genome is replicated. Certain environmental factors such as UV light can cause a switch from the lysogenic cycle to the lytic cycle. In the lytic cycle, new virus particles are made and released when the host cell lyses. Virulent phages are limited to just the lytic cycle.

- Archaeal Viruses
  - May be lytic or temperate
  - Most discovered so far are temperate by unknown mechanisms

B) Eukaryotic cells induction

4 scenarios:

Scenario 1: a cytocidal infection that causes the virus to multiply in the first cell lysing it and causing cell death ; then it will move on to adjacent cells

→Acute infection **figure 6.17 a**

Scenario 2: a persistent infection that lasts for years; the genetic material of the virus remains inside the cell without multiplying

→Latent infection **figure 6.17 b**

Scenario3: the virus enters & multiplies but it leaves the cell peacefully without lysis

→ Chronic infection **figure 6.17 c**

❖ Eukaryotic viruses can cause microscopic or macroscopic degenerative changes or abnormalities in host cells and in tissues that are distinct from lysis. These are called **cytopathic effects (CPEs)**.

Scenario 4: Transformation to malignant cell **figure 6.17 d**

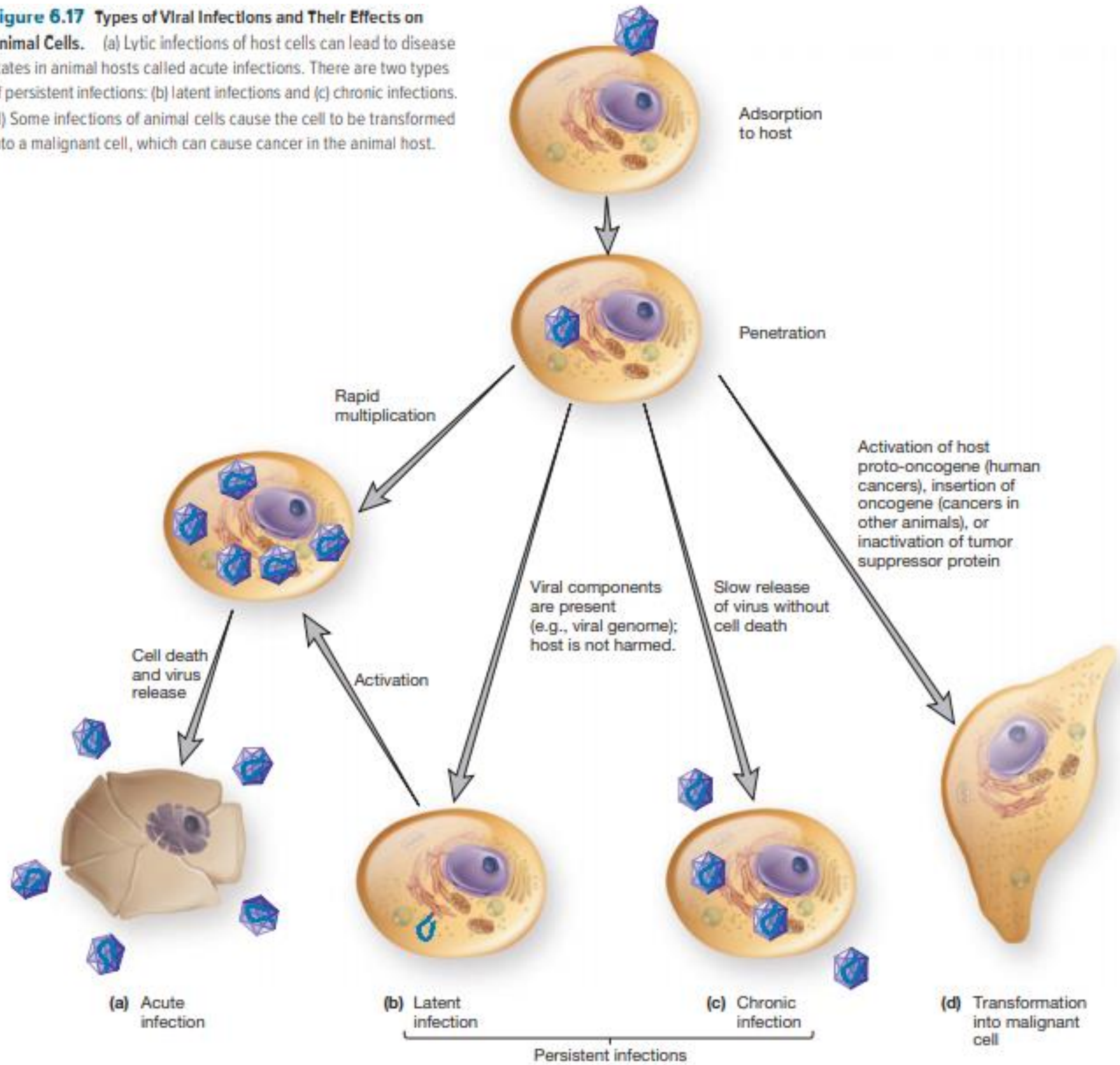



**Figure 6.17 Types of Viral Infections and Their Effects on Animal Cells.** (a) Lytic infections of host cells can lead to disease states in animal hosts called acute infections. There are two types of persistent infections: (b) latent infections and (c) chronic infections. (d) Some infections of animal cells cause the cell to be transformed into a malignant cell, which can cause cancer in the animal host.

- Viruses and Cancer
  - • Tumor
  - – growth or lump of tissue;
  - – benign tumors remain in place
  - • Neoplasia – abnormal new cell growth and reproduction due to loss of regulation
  - • Anaplasia – reversion to a more primitive or less differentiated state
  - • Metastasis – spread of cancerous cells throughout body
- Carcinogenesis: it is a complex , multistep process that involves oncogenes.

  Oncogenes are "cancer causing cells" that may come from a virus OR may be transformed host proto-oncogenes (الي همة اصلا بكونوا موجودين بالخلية المضيفة بس بدهم محفّز) (involved in normal regulation of cell growth/differentiation)

| Table 6.1 | Some Viruses Associated with Human Cancers | |
|---|---|---|
| **Virus** | **Genome Type** | **Cancer** |
| Human herpesvirus 8 (HHV8) | Double-stranded (ds) DNA | Several, including Kaposi's sarcoma |
| Epstein-Barr virus (EBV) | dsDNA | Several, including Burkitt's lymphoma and nasopharyngeal carcinoma |
| Hepatitis B virus | dsDNA | Hepatocellular carcinoma |
| Hepatitis C virus | Single-stranded (ss) RNA | Liver cancer |
| Human papillomaviruses (HPV) strains 6, 11, 16, and 18 | dsDNA | Cervical cancer and oral pharyngeal cancer |
| Human T-cell lymphotropic virus1 (HTLV-1) | ssRNA (retrovirus) | T-cell leukemia |

How can viruses cause cancer? (Possible Mechanisms by Which Viruses Cause Cancer)

- Viral proteins bind host cell tumor suppressor proteins
- Carrying an oncogene into cell and insert it into host genome
- Altered cell regulation
- Insertion of promoter or enhancer next to cellular oncogene

## Cultivation & Enumeration of viruses

First, it requires inoculation of <u>appropriate living host</u>

لانو الفايروس ما بتضاعف لحالو ولازم يكون عندو host cell ليعيش فاحنا ما رح نقدر ننميه ونزرعو غير بزراعة ال host المناسبة الو معو لهيك اهم اشي انو نعرف شو نوع هدول الخلايا cells

- Hosts for Bacterial and Archael Viruses
  - Usually we cultivate them in broth or agar cultures of suitable, young, actively growing bacteria
    →when viruses are grown in broth cultures, the evidence on its growing is the loss of i turbidity in the culture as viruses reproduce
    →when viruses are grown on agar, we will start seeing plaques of viruses
    (plaques are localized areas of cellular lysis and destruction that enlarge as the virus replicates )

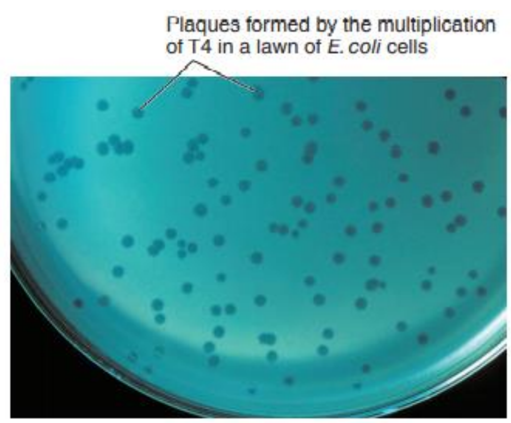



**Figure 6.18 Viral Plaques.** The plaques are formed in the bacterial cell lawn by the process illustrated in figure 6.20.

- Hosts for Animal Viruses
  - Tissue (cell) cultures: cells are infected with virus and then plaques are observed
  - Cytopathic effects (CPEs): microscopic or macroscopic degenerative changes or abnormalities in host cells and tissues

- Embryonated eggs: they are incubated at 37˚ and then viruses are purified from the liquid
- Hosts for Plant Viruses
  - Plant tissue cultures
  - Plant protoplast (الخلايا النباتية الي بتكون بدون جدار خلوي) cultures
  - Suitable whole plants: may cause localized <u>necrotic lesions</u> or generalized symptoms of infection

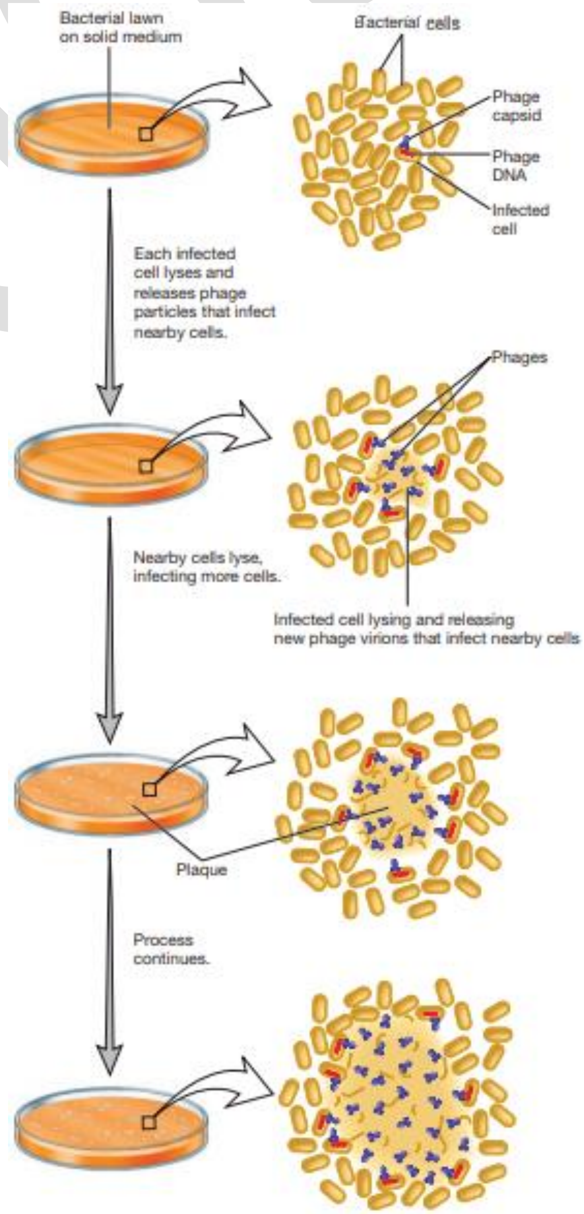
**Figure 6.19** **Necrotic Lesions on Plant Leaves.** A tobacco plant leaf showing leaf color changes caused by TMV.

- Quantification of Virus
  Can be:
  - Direct counting of viral particles

مناخد عينة ومنخففها بعدين منشوفها تحت الmicroscope

  - Or Indirect counting by an observable effect of the virus:
    - hemagglutination assay

منحط الفايروسات مع خلايا الدم الحمراء فبترتبط على سطحها. ازا كانت نسبة الفايروسات للخلايا كبيرة بشكل كافي، الvirions رح يربطوا خلايا الدم الحمراء مع بعض (او to agglutinate ) وبشكلوا شبكة بتوقفهم عن عملهم




    - plaque assays: dilutions of virus preparation made and plated on lawn of host cells and then the number of plaques is counted
      → Theoretically, when the multiplicity of infection is very low, then each plaque in a layer of host cells should have arisen from the multiplication of a single virion

      • results expressed as plaque-forming units (PFU) – plaque forming units (PFU) – PFU/ml = number of plaques/sample dilution
- Measuring Biological Effects:
  Infectious dose and lethal dose assay
  - Infectious dose (ID): is the dose that causes 50% of the host organisms to become infected
  - Lethal dose (LD): is the dilution that contains a concentration (dose) of virions large enough to destroy 50% of the host cells or organisms.
    Figure 6.21

## ❁ Viroids, Satellites & Proins

- Viroids: Infectious agents composed of closed, circular **single stranded RNAs** (ssRNAs) that do not encode gene products and their replication requires host cell DNA-dependent RNA polymerase .<u>They cause plant diseases</u> Figure 6.22 & 6.23
- Satellites:
  - Infectious nucleic acids (DNA or RNA)

    →3 types:

    1. satellite viruses encode their own capsid proteins when helped by a helper virus
    2. satellite RNAs
    3. satellite DNAs

       2& 3 do NOT encode their own capsid proteins
  - Encode one or more gene products
  - Require a helper virus for replication like:
    - human hepatitis D virus is satellite and requires human hepatitis B virus

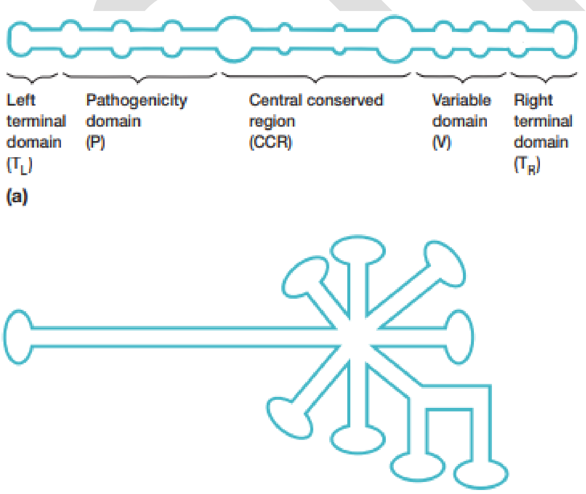



**Figure 6.23 Viroid Structure.** (a) This schematic diagram shows the general organization of a viroid belonging to the family *Pospiviroidae*. The closed single-stranded RNA circle has extensive intrastrand base pairing and interspersed unpaired loops. Also shown are the five domains identified in the molecule. Most changes in viroid pathogenicity arise from variations in the P and $T_L$ domains. (b) Schematic diagram of a viroid belonging to *Avsunviroidae*. These viroids lack the central conserved region observed in pospiviroids.

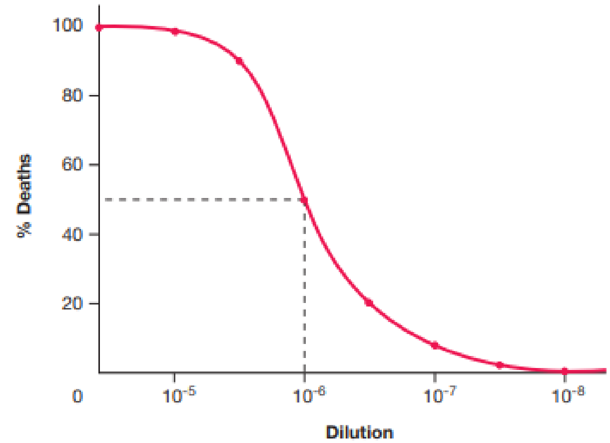



**Figure 6.21 Determination of $LD_{50}$.** The $LD_{50}$ is indicated by the dashed line.

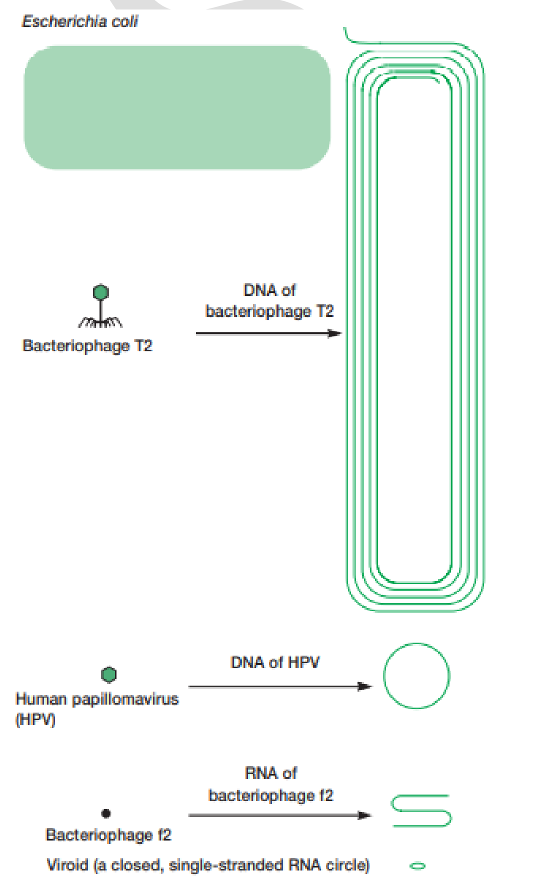



**Figure 6.22 Viroids, Viruses, and Bacteria.** A comparison of *E. coli*, several viruses, and the potato spindle-tuber viroid with respect to size and the amount of nucleic acid possessed. (All dimensions are enlarged approximately ×40,000.)

- Prions: they are proteinaceous(بروتيني) infectious particles that cause a variety of degenerative diseases in humans and animals mainly in the nervous system

  Diseases caused by prions:

  - **scrapie** in sheep
  - **bovine spongiform encephalopathy (BSE)** or mad cow disease: caused by feeding cows animal processed food instead of fodder
  - **Creutzfeldt-Jakob disease (CJD)** and **variant CJD (vCJD)** in humans
  - **kuru** in humans: is found among people from New Guinea who ate the brains of dead people as part of a funeral ritual

    →prions are highly stable, they can't be simply degraded using detergents or high temperatures and in order to degrade them, we use incinerators that have a temperature of 400°C !

The End

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